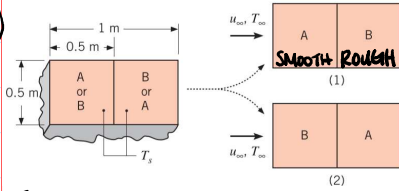


Homework 7

Thursday, February 27, 2020 2:26 PM

7.17)



$$T_s = 10^\circ\text{C} \quad T_\infty = 35^\circ\text{C}$$

$$U_\infty = 25 \text{ m/s}$$

To minimize total convection heat transfer 1/2?
Rate of convective heat transfer?

$$T_f = \frac{T_s + T_\infty}{2} \rightarrow T_f = \frac{10 + 35}{2} = 22.5^\circ\text{C} = 295.05 \text{ K}$$

Air properties @ 295.05 K from interpolation:

$$\nu: \frac{\nu - 11.44}{295.05 - 250} = \frac{15.89 - 11.44}{300 - 250} \rightarrow \nu = 15.50 \times 10^{-6} \text{ m}^2/\text{s}$$

$$k: \frac{k - 22.3}{295.05 - 250} = \frac{26.3 - 22.3}{300 - 250} \rightarrow k = 25.952 \times 10^{-3} \text{ W/m}\cdot\text{K}$$

$$Pr: \frac{Pr - 0.720}{295.05 - 250} = \frac{.707 - .720}{300 - 250} \rightarrow Pr = .708$$

$$\begin{aligned} \bar{Nu}_L &= C Re_L^m Pr^n \\ \bar{Sh}_L &= C Re_L^m Sc^n \end{aligned}$$

In orientation (1) the air flows through the smooth portion first this causes a laminar flow & turbulent flow in the second portion. Re will be low then high
 $\therefore$ total convection heat is lower

Convective heat transfer:

$$q = \bar{h}_L A_s (T_s - T_\infty) = 59.01 (.5) (10 - 35) \rightarrow \boxed{q = -737.59 \text{ W}}$$

$$\bar{h}_L = \frac{\bar{Nu}_L k}{L} = (2273.698) (25.952 \times 10^{-3}) = 59.01 \text{ W/m}^2\text{K}$$

$$\bar{Nu}_L = (.037 Re_L^{4/5} - 871) (Pr^{1/3}) = .037 (1012903.23)^{4/5} - 871 (.708)^{1/3} = 2273.698$$

$$Re_L = \frac{U_\infty L}{\nu} = \frac{25(1)}{15.5 \times 10^{-6}} = 1012903.23$$

7.38) $D = 100 \text{ mm} = .1 \text{ m}$ $L = 2 \text{ m}$ $T_s = 70^\circ\text{C}$ $V = 3 \text{ m/s}$ $T_\infty = 20^\circ\text{C}$

$$\bar{h}_L? \quad R_{th}? \quad q?$$

$$T_f = \frac{T_s + T_\infty}{2} \rightarrow T_f = 45^\circ\text{C} = 318 \text{ K}$$

Air Properties @ 318 K (SAME CALCS AS ABOVE)

$$k = .02703 \text{ W/m}\cdot\text{K}$$

$$\nu = 1.770 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = .7045$$

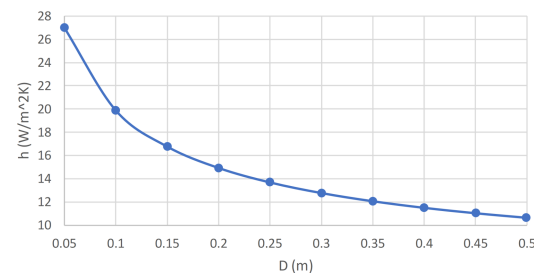
$$Re = \frac{VD}{\nu} = \frac{(3)(.1)}{1.77 \times 10^{-5}} = 16949.15$$

$$Nu = 0.3 + \frac{0.62 Re^{1/2} Pr^{1/3}}{[1 + (0.4/Pr)^{1/4}]^{1/4}} \left[1 + \left(\frac{Re}{282000} \right)^{5/8} \right]^{4/5} = 71.89$$

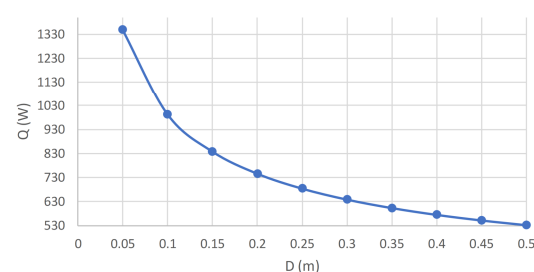
$$h = \frac{Nu k}{D} = \frac{(71.89)(.02703)}{.1} \rightarrow \boxed{h = 19.803 \text{ W/m}^2\text{K}}$$

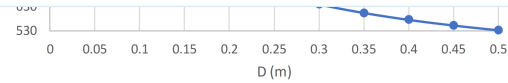
$$R_{th} = \frac{1}{h} = \frac{1}{19.803} = 0.0505 \text{ K/W} \rightarrow \boxed{R_{th} = .08012 \text{ K/W}}$$

Heat Transfer Coefficient with Varying Diameter



Heat Rate with Varying Diameter





$$R_{th} = \frac{1}{h A_s} = \frac{1}{h \pi D L} = \frac{1}{19.8637 (.1)(2)} \rightarrow \boxed{R_{th} = .08012 \text{ K/W}}$$

$$q = \frac{(T_s - T_{\infty})}{R_{th}} = \frac{70 - 20}{.08012} \rightarrow \boxed{q = 624.06 \text{ W}}$$

8.25) $\dot{m} = 1 \text{ Kg/s}$ $D = 5 \text{ mm} = .005 \text{ m}$ $T_i = 45^\circ\text{C}$ $T_o = 80^\circ\text{C}$ $T_s = 150^\circ\text{C}$
 $L = ?$ $= 318 \text{ K}$ $= 353 \text{ K}$ $= 423 \text{ K}$

Engine oil properties from interpolation:

$$\mu_i: \frac{\mu_i - 25.3}{318 - 310} = \frac{14.1 - 25.3}{320 - 310} \rightarrow \mu_i = 10.3 \times 10^{-2} \text{ Ns/m}^2$$

$$\mu_o: \frac{\mu_o - 3.56}{353 - 350} = \frac{2.52 - 3.56}{360 - 350} \rightarrow \mu_o = 3.25 \times 10^{-2} \text{ Ns/m}^2$$

$$Re_{di} = \frac{4\dot{m}_i}{\pi D \mu_i} = \frac{4}{\pi (.005)(10.3 \times 10^{-2})} = 1502.3 \therefore \text{LAMINAR FLOW}$$

$$Re_{do} = \frac{4\dot{m}_o}{\pi D \mu_o} = \frac{4}{\pi (.005)(3.25 \times 10^{-2})} = 7835.32 \therefore \text{TURBULENT FLOW}$$

Mean temp in laminar region:

$$\bar{T}_{m1} = \frac{45 + 52}{2} = 48.5^\circ\text{C} = 321.5 \text{ K}$$

Engine oil properties @ 321.5 K from interpolation:

$$\mu = 13.2 \times 10^{-2} \text{ Ns/m}^2$$

$$k = .143 \text{ W/m}\cdot\text{K}$$

$$Pr = 1850$$

$$C_p = 1999 \text{ J/Kg}\cdot\text{K}$$

$$Re_D = \frac{4\dot{m}}{\pi D \mu} = \frac{4}{\pi (.005)(13 \times 10^{-2})} = 1929.15$$

$$\bar{h} L_1 = -\frac{\dot{m} C_p}{\pi D} \ln \left[\frac{T_s - T_{m1}}{T_s - T_i} \right] = -\frac{1999}{\pi (.005)} \ln \left[\frac{150 - 52}{150 - 45} \right] = 8780$$

$$\bar{Nu}_{D1} = \frac{\bar{h} D}{k} = 3.66 + \frac{.0668 (D/L_1) Re_D Pr}{1 + .04 [(D/L_1) Re_D Pr]^{1/3}}$$

$$\rightarrow \frac{8780 D}{L k} \rightarrow 8780 = 3.66 \frac{K L_1}{D} + \frac{.0668 (D/L_1) Re_D Pr}{1 + .04 [(D/L_1) Re_D Pr]^{1/3}} \frac{K L_1}{D}$$

Assume $\bar{Nu}_{D1} = 10.9$

$$10.9 = \frac{\bar{h} D}{k} \rightarrow \bar{h} = 483.34 \rightarrow L = 18.1 \text{ m}$$

After calcs correct assumption ✓

Mean temp in Turbulent Region:

$$\bar{T}_{m2} = \frac{52 + 80}{2} = 66^\circ\text{C} = 339 \text{ K}$$

Engine oil properties @ 339 K

$$C_p = 2072 \text{ J/Kg}\cdot\text{K}$$

$$\mu = 5.02 \times 10^{-2} \text{ Ns/m}^2$$

$$k = .139 \text{ W/mK}$$

$$Pr = 834$$

$$Re_{D2} = \frac{4\dot{m}}{\pi D \mu} = \frac{4}{\pi (0.005)(5.02 \times 10^{-2})} = 4531.1$$

$$\text{friction factor} \rightarrow f = (79 \ln Re_{D2} - 1.64)^{-2} = .0398$$

$$Nu_{D2} = \frac{(f/8)(Re_{D2} - 1000) Pr}{1 + 12.7(f/8)^{1/4}(Pr^{1/4} - 1)} = 184$$

$$\bar{h} = \frac{Nu_{D2} k}{D} = \frac{(184)(.139)}{0.005} = 5120 \text{ W/m}^2\text{K}$$

$$L_2 = \frac{m c_p \ln[T_s - T_o]}{\pi D \bar{h} (T_s - T_{mt})} = 8.7 \text{ m}$$

$$L = L_1 + L_2 = 18.1 + 8.7 \rightarrow \boxed{L = 26.8 \text{ m}}$$

8.66) Laminar flow, constant T_s

a  Which rectangular channel $b/a = 1.0, 1.43, \text{ or } 2.0$ will give highest heat rate?

For $b/a = 1$

$$Nu = 2.98 = \frac{h D_h}{k} \rightarrow D_h = \frac{4 A_c}{P} = \frac{4ab}{2(a+b)} = \frac{4/a}{4/a} = a$$

$$h = \frac{2.98 K}{a}$$

For $b/a = 1.43$

$$Nu = 3.08 = \frac{h D_h}{k} \rightarrow D_h = \frac{4 A_c}{P} = \frac{4ab}{2(a+b)} = \frac{4a(1.43a)}{2(a+1.43a)} = \frac{5.72 a^2}{4.86 a} = 1.177 a$$

$$h = \frac{3.08 K}{1.177 a} \rightarrow h = \frac{2.617 K}{a}$$

For $b/a = 2.0$

$$Nu = 3.39 = \frac{h D_h}{k} \rightarrow D_h = \frac{4 A_c}{P} = \frac{4ab}{2(a+b)} = \frac{4a(2a)}{2(a+2a)} = \frac{8a^2}{6a} = 1.333 a$$

$$h = \frac{3.39 K}{1.33 a} \rightarrow h = \frac{2.543 K}{a}$$

$q = h A_s (T_s - T_{\infty}) \rightarrow$ greatest q corresponds to greatest h

$\hookrightarrow h = 2.98 K/a$ IS GREATEST

$\therefore b/a = 1$ will provide highest heat transfer rate

For circular tube:

$$Nu = 3.66 = \frac{h D}{k} \rightarrow h = \frac{3.66 K}{a} > \frac{2.98 K}{a} \therefore \text{Heat transfer rate is greater for circular tube}$$